Matrices

114. Let $A=\begin{bmatrix}2 & b & 1\\ b & b^2+1 & b\\ 1 & b & 2\end{bmatrix}$ where $b>0$. Then the minimum value of $\frac{\det(A)}{b}$ is :

(A) $\sqrt{3}$ (B) $-2\sqrt{3}$ (C) $-\sqrt{3}$ (D) $2\sqrt{3}$

माना $A=\begin{bmatrix}2 & b & 1\\ b & b^2+1 & b\\ 1 & b & 2\end{bmatrix}$ जहाँ $b>0$ है। तो $\frac{\det(A)}{b}$ का न्यूनतम मान है

(A) $\sqrt{3}$ (B) $-2\sqrt{3}$ (C) $-\sqrt{3}$ (D) $2\sqrt{3}$

Ans. D

Sol. $A=\begin{bmatrix}2 & b & 1\\ b & b^2+1 & b\\ 1 & b & 2\end{bmatrix}$

$\det(A)=2.\left[2b^2+2-b^2\right]-b(2b-b)+1\left[b^2-b^2-1\right]$

$= 2(b^2+2)-b(b)+1(-1)$

$= 2b^2+4-b^2-1$

$\det(A) = b^2+3$

Let $y=\frac{\det(A)}{b}=b+\frac{3}{b}$

Apply $AM \geq GM$

$\frac{b+\frac{3}{b}}{2}\geq\left(b\times\frac{3}{b}\right)^{1/2}$

$b+\frac{3}{b}\geq 2(3)^{1/2}$

115. If $d\in R$, and $A=\begin{bmatrix}-2 & 4+d & (\sin\theta)-2\\ 1 & (\sin\theta)+2 & d\\ 5 & (2\sin\theta)-d & (-\sin\theta)+2+2d\end{bmatrix}$, $\theta\in[0,2\pi]$. If the minimum value of det(A) is 8, then a value of d is :

(A) -7 (B) -5 (C) $2(\sqrt{2}+2)$ (D) $2(\sqrt{2}+1)$

माना $d\in R$ तथा $A=\begin{bmatrix}-2 & 4+d & (\sin\theta)-2\\ 1 & (\sin\theta)+2 & d\\ 5 & (2\sin\theta)-d & (-\sin\theta)+2+2d\end{bmatrix}$, $\theta\in[0,2\pi]$ यदि det(A) का न्यूनतम मान 8 है, तो d का एक मान

है:

(A) -7 (B) -5 (C) $2(\sqrt{2}+2)$ (D) $2(\sqrt{2}+1)$

Ans. B

Sol. $A=\begin{vmatrix}-2 & 4+d & \sin\theta-2\\ 1 & \sin\theta+2 & d\\ 5 & 2\sin\theta-d & -\sin\theta+2+2d\end{vmatrix}$

$R_1 \to R_1 + R_3 - 2R_2$

$$A(\theta)=\begin{vmatrix}1 & 0 & 0\\ 1 & \sin\theta+2 & d\\ 5 & 2\sin\theta-d & -\sin\theta+2+2d\end{vmatrix}$$

$A(\theta) = (\sin\theta + 2)(2 + 2d - \sin\theta) - \sin\theta + 2 + 2d$

$A(\theta) = (d+2)^2 - \sin^2\theta$

$A_{min} = (d+2)^2 - 1 = 8$

$d + 2 = \pm 3$

$d = 1, -5$

116. Let $P=\begin{bmatrix}1 & 0 & 0\\ 3 & 1 & 0\\ 9 & 3 & 1\end{bmatrix}$ and $Q=[q_{ij}]$ be two 3×3 matrices such that $Q - P^5 = I_3$. Then $\frac{q_{21}+q_{31}}{q_{32}}$ is equal to :

(A) 10 (B) 9 (C) 15 (D) 135

यदि $P=\begin{bmatrix}1 & 0 & 0\\ 3 & 1 & 0\\ 9 & 3 & 1\end{bmatrix}$ तथा $Q=[q_{ij}]$ दो ऐसे 3×3 आव्यूह हैं, कि $Q - P^5 = I_3$ है, तो $\frac{q_{21}+q_{31}}{q_{32}}$ बराबर है –

(A) 10 (B) 9 (C) 15 (D) 135

Ans. A

Sol. $P=\begin{bmatrix}1 & 0 & 0\\ 3 & 1 & 0\\ 9 & 3 & 1\end{bmatrix}$

$P^2=\begin{bmatrix}1 & 0 & 0\\ 3+3 & 1 & 0\\ 9+3\times3+1 & 3+3 & 1\end{bmatrix}$

$$P^5 = \begin{bmatrix} 1 & 0 & 0 \\ 15 & 1 & 0 \\ 135 & 15 & 1 \end{bmatrix}$$

$Q - P^5 = I_3$
$Q = I_3 + P^5$

$$= \begin{bmatrix} 2 & 0 & 0 \\ 15 & 2 & 0 \\ 135 & 15 & 2 \end{bmatrix}$$

$$\frac{q_{21} + q_{31}}{q_{32}} \Rightarrow 10$$

117. If $A = \begin{bmatrix} e^t & e^{-t}\cos t & e^{-t}\sin t \\ e^t & -e^{-t}\cos t - e^{-t}\sin t & -e^{-t}\sin t + e^{-t}\cos t \\ e^t & 2e^{-t}\sin t & -2e^{-t}\cos t \end{bmatrix}$ then A is :

(A) invertible only if $t = \pi$ (B) not invertible for any $t \in R$

(C) invertible for all $t \in R$ (D) invertible only if $t = \frac{\pi}{2}$

यदि $A = \begin{bmatrix} e^t & e^{-t}\cos t & e^{-t}\sin t \\ e^t & -e^{-t}\cos t - e^{-t}\sin t & -e^{-t}\sin t + e^{-t}\cos t \\ e^t & 2e^{-t}\sin t & -2e^{-t}\cos t \end{bmatrix}$ है तो A :

(A) व्युत्क्रमणीय है, केवल $t = \pi$ है। (B) किसी भी $t \in R$ के लिए व्युत्क्रमणीय नहीं है।

(C) सभी $t \in R$ के लिए व्युत्क्रमणीय है। (D) व्युत्क्रमणीय है, केवल तब, जब $t = \frac{\pi}{2}$ है।

Ans. C

Sol. $$|A| = e^{-t}\begin{vmatrix} 1 & \cos t & \sin t \\ 1 & -\cos t - \sin t & -\sin t + \cos t \\ 1 & 2\sin t & -2\cos t \end{vmatrix}$$

$$R_2 \to R_2 - R_1, R_3 \to R_3 - R_1 = \begin{vmatrix} 1 & \cos t & \sin t \\ 0 & -\cos t - \sin t & -2\sin t + \cos t \\ 0 & 2\sin t - \cos t & -2\cos t - \sin t \end{vmatrix}$$

$= (2\cos t + \sin t)^2 + (2\sin t - \cos t)^2$
$|A| = 5$
So A is invertible for all $t \in R$

118. Let $A=\begin{pmatrix}\cos\alpha & -\sin\alpha\\ \sin\alpha & \cos\alpha\end{pmatrix}$,$(\alpha\in R)$ such that $A^{32}=\begin{pmatrix}0 & -1\\ 1 & 0\end{pmatrix}$ then a value of α is

(A) $\frac{\pi}{32}$ (B) $\frac{\pi}{64}$ (C) 0 (D) $\frac{\pi}{16}$

माना $A=\begin{pmatrix}\cos\alpha & -\sin\alpha\\ \sin\alpha & \cos\alpha\end{pmatrix}$,$(\alpha\in R)$ इस प्रकार है कि $A^{32}=\begin{pmatrix}0 & -1\\ 1 & 0\end{pmatrix}$, तो α का एक मान है :

(A) $\frac{\pi}{32}$ (B) $\frac{\pi}{64}$ (C) 0 (D) $\frac{\pi}{16}$

Ans. B

Sol. $A=\begin{bmatrix}\cos\alpha & -\sin\alpha\\ \sin\alpha & \cos\alpha\end{bmatrix}$

$$A^2=\begin{bmatrix}\cos\alpha & -\sin\alpha\\ \sin\alpha & \cos\alpha\end{bmatrix}\begin{bmatrix}\cos\alpha & -\sin\alpha\\ \sin\alpha & \cos\alpha\end{bmatrix}=\begin{bmatrix}\cos 2\alpha & -\sin 2\alpha\\ \sin 2\alpha & \cos 2\alpha\end{bmatrix}$$

Then $A^4=A^2.A^2=\begin{bmatrix}\cos 4\alpha & -\sin 4\alpha\\ \sin 4\alpha & \cos 4\alpha\end{bmatrix}$

Similarly $A^8=A^4.A^4=\begin{bmatrix}\cos 8\alpha & -\sin 8\alpha\\ \sin 8\alpha & \cos 8\alpha\end{bmatrix}$

and so on $A^{32}=\begin{bmatrix}\cos 32\alpha & -\sin 32\alpha\\ \sin 32\alpha & \cos 32\alpha\end{bmatrix}=\begin{bmatrix}0 & -1\\ 1 & 0\end{bmatrix}$

So $\sin 32\alpha = 1$ and $\cos 32\alpha = 0$

$$\Rightarrow \quad 32\alpha=2n\pi+\frac{\pi}{2} \quad \Rightarrow \quad \alpha=\frac{n\pi}{16}+\frac{\pi}{64} \text{ where } n\in Z$$

put $n=0,\ \alpha=\frac{\pi}{64}$

119. Let the numbers 2, b, c be in an A.P. and $A=\begin{bmatrix}1 & 1 & 1\\ 2 & b & c\\ 4 & b^2 & c^2\end{bmatrix}$. If $\det(A)\in[2,16]$, then c lies in the interval :

(A) [2, 3) (B) $(2+2^{3/4}, 4)$ (C) $[3, 2+2^{3/4}]$ (D) [4, 6]

माना संख्याएं 2, b, c एक समान्तर श्रेढ़ी में हैं तथा $A=\begin{bmatrix}1 & 1 & 1\\ 2 & b & c\\ 4 & b^2 & c^2\end{bmatrix}$. यदि $\det(A)\in[2,16]$, तो c निम्न में से किस अन्तराल में है :

(A) [2, 3) (B) $(2+2^{3/4}, 4)$ (C) $[3, 2+2^{3/4}]$ (D) [4, 6]

Ans. D

Sol. $|A|=\begin{vmatrix}1 & 1 & 1\\ 2 & b & c\\ 4 & b^2 & c^2\end{vmatrix}$ $\quad c_2 \to c_2 - c_1,\ c_3 \to c_3 - c_1$

$$|A|=\begin{vmatrix}1 & 0 & 0\\ 2 & b-2 & c-2\\ 4 & (b-2)(b+2) & (c-2)(c+2)\end{vmatrix} = (b-2)(c-2)(c-b)$$

2, b, c are in A.P. $\Rightarrow$ $(b-2) = (c-b) = d,\ c-2 = 2d$

$\Rightarrow$ $|A| = d.2d.\ d = 2d^3$

$\because$ $|A| \in [2,16] \Rightarrow 1 \le d^3 \le 8 \Rightarrow 1 \le d \le 2$

$4 \le 2d + 2 \le 6 \Rightarrow 4 \le c \le 6$

120. If $\begin{bmatrix}1 & 1\\ 0 & 1\end{bmatrix}\cdot\begin{bmatrix}1 & 2\\ 0 & 1\end{bmatrix}\cdot\begin{bmatrix}1 & 3\\ 0 & 1\end{bmatrix}\cdot\ldots\cdot\begin{bmatrix}1 & n-1\\ 0 & 1\end{bmatrix}=\begin{bmatrix}1 & 78\\ 0 & 1\end{bmatrix}$, then the inverse of $\begin{bmatrix}1 & n\\ 0 & 1\end{bmatrix}$ is :

(A) $\begin{bmatrix}1 & 0\\ 13 & 1\end{bmatrix}$ (B) $\begin{bmatrix}1 & -13\\ 0 & 1\end{bmatrix}$ (C) $\begin{bmatrix}1 & -12\\ 0 & 1\end{bmatrix}$ (D) $\begin{bmatrix}1 & 0\\ 12 & 1\end{bmatrix}$

यदि $\begin{bmatrix}1 & 1\\ 0 & 1\end{bmatrix}\cdot\begin{bmatrix}1 & 2\\ 0 & 1\end{bmatrix}\cdot\begin{bmatrix}1 & 3\\ 0 & 1\end{bmatrix}\cdot\ldots\cdot\begin{bmatrix}1 & n-1\\ 0 & 1\end{bmatrix}=\begin{bmatrix}1 & 78\\ 0 & 1\end{bmatrix}$ है, तो $\begin{bmatrix}1 & n\\ 0 & 1\end{bmatrix}$ का व्युत्क्रम (inverse) है :

(A) $\begin{bmatrix}1 & 0\\ 13 & 1\end{bmatrix}$ (B) $\begin{bmatrix}1 & -13\\ 0 & 1\end{bmatrix}$ (C) $\begin{bmatrix}1 & -12\\ 0 & 1\end{bmatrix}$ (D) $\begin{bmatrix}1 & 0\\ 12 & 1\end{bmatrix}$

Ans. B

Sol. $\begin{bmatrix}1 & 1\\ 0 & 1\end{bmatrix}\begin{bmatrix}1 & 2\\ 0 & 1\end{bmatrix}\begin{bmatrix}1 & 3\\ 0 & 1\end{bmatrix}\begin{bmatrix}1 & 4\\ 0 & 1\end{bmatrix}\ldots\ldots\begin{bmatrix}1 & n-1\\ 0 & 1\end{bmatrix}=\begin{bmatrix}1 & 78\\ 0 & 1\end{bmatrix}$

$\Rightarrow$ $\begin{bmatrix}1 & 1+2+3+\ldots.+(n-1)\\ 0 & 1\end{bmatrix}=\begin{bmatrix}1 & 78\\ 0 & 1\end{bmatrix}$

$\Rightarrow$ $\dfrac{(n-1)n}{2} = 78$

$\Rightarrow$ $n = 13$

Now, inverse of $\begin{bmatrix}1 & 13\\ 0 & 1\end{bmatrix}=\begin{bmatrix}1 & -13\\ 0 & 1\end{bmatrix}$

121. The total number of matrices

$A=\begin{pmatrix} 0 & 2y & 1 \\ 2x & y & -1 \\ 2x & -y & 1 \end{pmatrix}$, $(x, y \in R, x \neq y)$ for

Which $A^TA = 3I_3$ is :

(A) 6 (B) 3 (C) 4 (D) 2

आव्यूहों $A=\begin{pmatrix} 0 & 2y & 1 \\ 2x & y & -1 \\ 2x & -y & 1 \end{pmatrix}$, $(x, y \in R, x \neq y)$

जिनके लिए $A^TA = 3I_3$ है, की कुल संख्या है :

(A) 6 (B) 3 (C) 4 (D) 2

Ans. C

Sol. $A^TA=\begin{bmatrix} 0 & 2x & 2x \\ 2y & y & -y \\ 1 & -1 & 1 \end{bmatrix}\begin{bmatrix} 0 & 2y & 1 \\ 2x & y & -1 \\ 2x & -y & 1 \end{bmatrix}=3I$

$=\begin{bmatrix} 8x^2 & 0 & 0 \\ 0 & 6y^2 & 0 \\ 0 & 0 & 3 \end{bmatrix}\begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$

$\Rightarrow \quad 8x^2 = 3, 6y^2 = 3$

$x=\pm\sqrt{\frac{3}{8}}, y=\pm\sqrt{\frac{1}{2}}$

Total combinations of $(x, y) = 2 \times 2 = 4$

122. Let $A = [a_{ij}]$ and $B = [b_{ij}]$ be two 3×3 real matrices such that $b_{ij} = (3)^{(i+j-2)}a_{ji}$, where i, j = 1, 2, 3. If the determinant of B is 81, then determinant of A is:

(A) 3 (B) 1/3 (C) 1/81 (D) 1/9

माना $A = [a_{ij}]$ तथा $B = [b_{ij}]$, 3×3 के दो वास्तविक आव्यूह इस प्रकार हैं कि $b_{ij} = (3)^{(i+j-2)}a_{ji}$, जहाँ i, j = 1, 2, 3. यदि B का सारणिक 81 है, तो A का सारणिक है :

(A) 3 (B) 1/3 (C) 1/81 (D) 1/9

Ans. D

Sol. $B = \begin{bmatrix} a_{11} & 3a_{21} & 3^2a_{31} \\ 3a_{12} & 3^2a_{22} & 3^3a_{32} \\ 3^2a_{13} & 3^3a_{23} & 3^4a_{33} \end{bmatrix}$

$|B| = 3^6|A| = 81$

$|A| = \frac{1}{9}$

123. Let $A = \{X = (x, y, z)^T : PX = 0 \text{ and } x^2 + y^2 + z^2 = 1\}$, where $P = \begin{bmatrix} 1 & 2 & 1 \\ -2 & 3 & -4 \\ 1 & 9 & -1 \end{bmatrix}$, then the set A :

(A) contains more than two elements. (B) contains exactly two elements.

(C) is a singleton. (D) is an empty set.

यदि $A = \{X = (x, y, z)^T : PX = 0$ तथा $x^2 + y^2 + z^2 = 1\}$ जबकि $P = \begin{bmatrix} 1 & 2 & 1 \\ -2 & 3 & -4 \\ 1 & 9 & -1 \end{bmatrix}$ है, तो A :

(A) में दो से अधिक अवयव हैं। (B) में मात्र दो अवयव हैं।

(C) एक ही अवयव वाला समुच्चय है। (D) एक रिक्त समुच्चय है।

Ans. B

Sol. $|P| = (-3 + 36) - 2(2 + 4) + 1(-18 - 3)$

$= 33 - 12 - 21 = 0$

$PX = 0$ has infinite solution

$x + 2y + z = 0$

$-2x + 3y - 4z = 0$

$x + 9y - z = 0$

put $z = \lambda$

$x + 2y = -\lambda$

$\frac{x + 9y = \lambda}{y = 2\lambda / 7}$

$x = \frac{-11\lambda}{7}$

$(x, y, z) \equiv \left(\frac{-11\lambda}{7}, \frac{2\lambda}{7}, \lambda\right)$

$x^2 + y^2 + z^2 = 1$

$$\lambda^2\left(\frac{121}{49}+\frac{4}{49}+1\right)=1$$

$$\lambda = \pm\frac{7}{\sqrt{174}}$$

Two set of values

124. Let $A=\begin{bmatrix} x & 1 \\ 1 & 0 \end{bmatrix}, x \in \mathbf{R}$ and $A^4 = [a_{ij}]$. If $a_{11} = 109$, then a_{22} is equal to _____

माना $A=\begin{bmatrix} x & 1 \\ 1 & 0 \end{bmatrix}, x \in \mathbf{R}$ तथा $A^4 = [a_{ij}]$ यदि $a_{11} = 109$ है, तो a_{22} बराबर है _____ ।

Ans. 10

Sol. $$A^2=\begin{bmatrix} x & 1 \\ 1 & 0 \end{bmatrix}.\begin{bmatrix} x & 1 \\ 1 & 0 \end{bmatrix}=\begin{bmatrix} x^2+1 & x \\ x & 1 \end{bmatrix}$$

$$A^4=\begin{bmatrix} x^2+1 & x \\ x & 1 \end{bmatrix}\begin{bmatrix} x^2+1 & x \\ x & 1 \end{bmatrix}$$

$$A^4=\begin{bmatrix} (x^2+1)^2+x^2 & ____ \\ ____ & x^2+1 \end{bmatrix}=\begin{bmatrix} 109 & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

$(x^2 + 1)^2 + x^2 = 109 \Rightarrow x^2 = 9$

$a_{22} = x^2 + 1 = 10$

125. Let A be a 3 × 3 matrix such that adj $A=\begin{bmatrix} 2 & -1 & 1 \\ -1 & 0 & 2 \\ 1 & -2 & -1 \end{bmatrix}$ and B = adj(adj A).

If $|A| = \lambda$ and $|(B^{-1})^T| = \mu$, then the ordered pair, $(|\lambda|, \mu)$ is equal to :

माना A एक 3 × 3 आव्यूह है, जिसके लिए $A=\begin{bmatrix} 2 & -1 & 1 \\ -1 & 0 & 2 \\ 1 & -2 & -1 \end{bmatrix}$ तथा B = adj(adj A) हैं। यदि If $|A| = \lambda$ तथा $|(B^{-1})^T| = \mu$ हैं, तो क्रमित युग्म $(|\lambda|, \mu)$ बराबर है :

(A) (3, 81) (B) $\left(9,\frac{1}{9}\right)$ (C) $\left(3,\frac{1}{81}\right)$ (D) $\left(9,\frac{1}{81}\right)$

Ans. C

Sol. $|\text{adj } A| = \begin{vmatrix} 2 & -1 & 1 \\ -1 & 0 & 2 \\ 1 & -2 & -1 \end{vmatrix} = |A|^{n-1}$

$\therefore |A|^2 = 9$

$|A| = \pm 3$

$|A| = \lambda = \pm 3$

$|\lambda| = 3$

$B = \text{adj (adj } A)$

$|B| = |\text{adj (adj } A)|$

$|B| = |A|^{(n-1)^2}$

$|B| = |A|^4$

$|B| = 81$

$\left|\left(B^{-1}\right)^T\right| = \mu$

$\left|B^{-1}\right| = \mu$

$\frac{1}{|B|} = \mu$

$\mu = \frac{1}{81}$

$= \left(3, \frac{1}{81}\right)$

126. If $A = \begin{bmatrix} \cos\theta & i\sin\theta \\ i\sin\theta & \cos\theta \end{bmatrix}, \left(\theta = \frac{\pi}{24}\right)$ and $A^5 = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, where $i = \sqrt{-1}$, then which one of the following is not true?

यदि $A = \begin{bmatrix} \cos\theta & i\sin\theta \\ i\sin\theta & \cos\theta \end{bmatrix}, \left(\theta = \frac{\pi}{24}\right)$ तथा $A^5 = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, जबकि $i = \sqrt{-1}$ है, तो निम्न में से कौन–सा सत्य नहीं है ?

(A) $a^2 - d^2 = 0$ (B) $a^2 - b^2 = \frac{1}{2}$ (C) $0 \le a^2 + b^2 \le 1$ (D) $a^2 - c^2 = 1$

Ans. B

Sol. $A^2 = \begin{bmatrix} \cos\theta & i\sin\theta \\ i\sin\theta & \cos\theta \end{bmatrix}\begin{bmatrix} \cos\theta & i\sin\theta \\ i\sin\theta & \cos\theta \end{bmatrix}$

$= \begin{bmatrix} \cos^2\theta + i^2\sin^2\theta & i(2\cos\theta\sin\theta) \\ 2i\sin\theta.\cos\theta & \cos^2\theta + i^2\sin^2\theta \end{bmatrix} = \begin{bmatrix} \cos 2\theta & i\sin 2\theta \\ i\sin 2\theta & \cos 2\theta \end{bmatrix}$

$A^3 = \begin{bmatrix} \cos 2\theta & i\sin 2\theta \\ i\sin 2\theta & \cos 2\theta \end{bmatrix}\begin{bmatrix} \cos\theta & i\sin\theta \\ i\sin\theta & \cos\theta \end{bmatrix}$

$= \begin{bmatrix} \cos 3\theta & i\sin 3\theta \\ i\sin 3\theta & \cos 3\theta \end{bmatrix}$

$A^5 = \begin{bmatrix} \cos 5\theta & i\sin 5\theta \\ i\sin 5\theta & \cos 5\theta \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$

$a = \cos 5\theta$ $b = i\sin 5\theta$

$c = i\sin 5\theta$ $d = \cos 5\theta$

$a^2 - b^2 = \cos^2 5\theta - i^2\sin^2 5\theta = \cos^2 5\theta + \sin^2 5\theta = 1$

$a^2 - d^2 = \cos^2 5\theta - \cos^2 5\theta = 0$

$a^2 + b^2 = \cos^2 5\theta - \sin^2 5\theta = \cos 10\theta = \cos\frac{5\pi}{12} \in (0,1)$

$a^2 - c^2 = \cos^2 5\theta - i^2\sin^2 5\theta = \cos^2 5\theta + \sin^2 5\theta = 1$

127. Suppose the vectors x_1, x_2 and x_3 are the solutions of the system of linear equations, $Ax = b$ when the vector b on the right side is equal to b_1, b_2 and b_3 respectively. If

$x_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, x_2 = \begin{bmatrix} 0 \\ 2 \\ 1 \end{bmatrix}, x_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, b_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, b_2 = \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix}$ and $b_3 = \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix}$, then the determinant of A is equal to :

माना सदिश x_1, x_2 तथा x_3, रैखिक समीकरण निकाय $Ax = b$ के हल हैं, जबकि दांई ओर का सदिश b, क्रमशः b_1, b_2 तथा b_3 के बराबर है। यदि

$x_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, x_2 = \begin{bmatrix} 0 \\ 2 \\ 1 \end{bmatrix}, x_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, b_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, b_2 = \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix}$ तथा $b_3 = \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix}$ हैं, तो A के सारणिक का मान है :

(A) $\frac{3}{2}$ (B) 2 (C) 4 (D) $\frac{1}{2}$

Ans. B

Sol. Let $X = [x_1\ x_2\ x_3] = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 2 & 0 \\ 1 & 1 & 1 \end{bmatrix}$

$$B = [b_1\ b_2\ b_3] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

$AX = B$

$\Rightarrow |A|\ |X| = |B|$

$\Rightarrow |A| = \frac{|B|}{|X|} = \frac{4}{2} = 2$

128. Let $\theta = \frac{\pi}{5}$ and $A = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$. If $B = A + A^4$, then det (B) :

माना $\theta = \frac{\pi}{5}$ तथा $A = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$ है। यदि If $B = A + A^4$, तो det (B) :

(A) is one (B) lies in (1, 2) (C) lies in (2, 3) (D) is zero

(A) is के बराबर (B) अंतराल (1, 2) में है। (C) अंतराल (2, 3) में है। (D) zero के बराबर है।

Ans. B

Sol. $B = A(I + A^3)$

$|B| = |A|\,|I + A^3|$

$|A| = 1$

$|B| = |I + A^3|$

$$A^2 = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$$

$$A^3 = \begin{bmatrix} \cos 2\theta & \sin 2\theta \\ -\sin 2\theta & \cos 2\theta \end{bmatrix} \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$$

$$= \begin{bmatrix} \cos 3\theta & \sin 3\theta \\ -\sin 3\theta & \cos 3\theta \end{bmatrix}$$

$$A^3 + I = \begin{bmatrix} 1+\cos 3\theta & \sin 3\theta \\ -\sin 3\theta & 1+\cos 3\theta \end{bmatrix}$$

$|A^3 + I| = (1 + \cos 3\theta)^2 + \sin^2 3\theta$

$|B| = 1 + 2\cos 3\theta + 1 = 2(1 + \cos 3\theta)$

$|B| = 2(1 + \cos 108^\circ) = 2(1 - \cos 72^\circ) = 2(1 - \sin 18^\circ)$

$$= 2\left(1 - \frac{\sqrt{5}-1}{4}\right) = 2\left(\frac{5-\sqrt{5}}{4}\right) = \left(\frac{5-\sqrt{5}}{2}\right)$$

129. Let A and B be 3 × 3 real matrices such that A is symmetric matrix and B is skew-symmetric matrix. Then the system of linear equations $(A^2B^2 - B^2A^2)X = O$, where X is a 3 × 1 column matrix of unknown variables and O is a 3 × 1 null matrix, has :

(A) no solution (B) exactly two solutions

(C) infinitely many solutions (D) a unique solution

माना A तथा B दो 3 × 3 वास्तिवक आव्यूह हैं जबकि A सममित आव्यूह है तथा B विषम सममित आव्यूह है। तो रैखिक समीकरण निकाय, $(A^2B^2 - B^2A^2)X = O$, जबकि X एक 3 × 1 अज्ञात चरों का स्तम्भ आव्यूह है तथ O एक 3 × 1 शून्य आव्यूह है :

(A) के ठीक दो हल हैं (B) का कोई भी हल नहीं है

(C) का केवल एक हल है (D) के अनन्त हल हैं

Ans. C

Sol. $(A^2B^2 - B^2A^2)^T = (A^2B^2)^T - (B^2A^2)^T$

$= B^2A^2 - A^2B^2$

$\Rightarrow A^2B^2 - B^2A^2$ is skew – symmetic

$\Rightarrow |A^2B^2 - B^2A^2| = 0$

$\Rightarrow$ option (3)

130. If $A = \begin{bmatrix} 0 & -\tan\left(\frac{\theta}{2}\right) \\ \tan\left(\frac{\theta}{2}\right) & 0 \end{bmatrix}$ and $(I_2 + A)(I_2 - A)^{-1} = \begin{bmatrix} a & -b \\ b & a \end{bmatrix}$, then $13(a^2 + b^2)$ is equal to _____.

यदि $A=\begin{bmatrix} 0 & -\tan\left(\frac{\theta}{2}\right) \\ \tan\left(\frac{\theta}{2}\right) & 0 \end{bmatrix}$ तथा $(I_2+A)(I_2-A)^{-1}=\begin{bmatrix} a & -b \\ b & a \end{bmatrix}$ हैं, तो $13(a^2+b^2)$ बराबर है :

Ans. 13

Sol. $A=\begin{bmatrix} 0 & -\tan\theta \\ \tan\frac{\theta}{2} & 0 \end{bmatrix}$

$$(I_2+A)(I_2-A)^{-1}=\begin{bmatrix} a & -b \\ b & a \end{bmatrix}$$

$$(I_2+A)=\begin{bmatrix} a & -b \\ b & a \end{bmatrix}(I_2-A)$$

$$\begin{bmatrix} 1 & -\tan\frac{\theta}{2} \\ \tan\frac{\theta}{2} & 1 \end{bmatrix}=\begin{bmatrix} a & -b \\ b & a \end{bmatrix}\begin{bmatrix} 1 & \tan\frac{\theta}{2} \\ -\tan\frac{\theta}{2} & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -\tan\frac{\theta}{2} \\ \tan\frac{\theta}{2} & 1 \end{bmatrix}=\begin{bmatrix} a+b\tan\frac{\theta}{2} & a\tan\frac{\theta}{2}-b \\ b-a\tan\frac{\theta}{2} & b\tan\frac{\theta}{2}+a \end{bmatrix}$$

By comparison

$$a+b\tan\frac{\theta}{2}=1 \quad ...(1)$$

$$a\tan\frac{\theta}{2}-b=-\tan\frac{\theta}{2} \quad ...(2)$$

$$b-a\tan\frac{\theta}{2}=\tan\frac{\theta}{2} \quad ...(3)$$

$$b\tan\frac{\theta}{2}+a=1 \quad ...(4)$$

By, $(1)^2+(2)^2$

$a^2+b^2=1$

$13(a^2+b^2)=13$

131. Let A be a symmetric matrix of order 2 with integer entries. If the sum of the diagonal elements of A^2 is 1, then the

possible number of such matrices is :

माना A एक 2 कोटि का सममित आव्यूह है, जिसके अवयव पूर्णांक हैं। यदि A^2 के विकर्ण के अवयवों का योगफल 1 है, तो ऐसे आव्यूहों की संभावित संख्या है :

(A) 1 (B) 12 (C) 6 (D) 4

Ans. D

Sol. Let $A=\begin{bmatrix} a & b \\ b & c \end{bmatrix}$; $a, b, c, d \in$ Integer.

$$A^2=\begin{bmatrix} a & b \\ b & c \end{bmatrix}\begin{bmatrix} a & b \\ b & c \end{bmatrix}=\begin{bmatrix} a^2+b^2 & ab+bc \\ ab+bc & b^2+c^2 \end{bmatrix}$$

Sum of diagonal entries $= a^2+b^2+b^2+c^2=1$

$$\Rightarrow a^2+2b^2+c^2=1$$

$$\Rightarrow a=\pm1,\ b=0=C$$

O R $a=b=0,\ c=\pm1.$

$\Rightarrow$ 4 possible matrices

132. If $A=\begin{pmatrix} 0 & \sin\alpha \\ \sin\alpha & 0 \end{pmatrix}$ and $\det\left(A^2-\frac{1}{2}I\right)=0$, then a possible value of α is :

यदि $A=\begin{pmatrix} 0 & \sin\alpha \\ \sin\alpha & 0 \end{pmatrix}$ तथा $\det\left(A^2-\frac{1}{2}I\right)=0$ है, तो α का एक संभव मान है :

(A) $\frac{\pi}{3}$ (B) $\frac{\pi}{6}$ (C) $\frac{\pi}{4}$ (D) $\frac{\pi}{2}$

Ans. C

Sol. $$A^2=\begin{bmatrix} 0 & \sin\alpha \\ \sin\alpha & 0 \end{bmatrix}\begin{bmatrix} 0 & \sin\alpha \\ \sin\alpha & 0 \end{bmatrix}$$

$$A^2=\begin{bmatrix} \sin^2\alpha & 0 \\ 0 & \sin^2\alpha \end{bmatrix}$$

$$A^2-\frac{I}{2}=\begin{bmatrix} \sin^2\alpha-\frac{1}{2} & 0 \\ 0 & \sin^2\alpha-\frac{1}{2} \end{bmatrix}=\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\left(\sin^2\alpha-\frac{1}{2}\right)^2=0$$

$\sin\alpha=\frac{1}{\sqrt{2}}$, $\sin\alpha=\frac{-1}{\sqrt{2}}$

Rejected

$\alpha=\frac{\pi}{4}$ $\alpha\in\left[0\quad\frac{\pi}{2}\right]$

133. Let $A=\begin{bmatrix} i & -i \\ -i & i \end{bmatrix}$, $i=\sqrt{-1}$. Then the system of linear equations $A^8\begin{bmatrix} x \\ y \end{bmatrix}=\begin{bmatrix} 8 \\ 64 \end{bmatrix}$ has:

(A) Infinitely many solutions (B) Exactly two solutions

(C) A unique solution (D) No solution

माना $A=\begin{bmatrix} i & -i \\ -i & i \end{bmatrix}$, $i=\sqrt{-1}$ हैं। तो रैखिक समीकरण निकाय $A^8\begin{bmatrix} x \\ y \end{bmatrix}=\begin{bmatrix} 8 \\ 64 \end{bmatrix}$:

(A) का कोई नहीं है (B) के अनंत हल हैं

(C) के मात्र दो हल हैं (D) का अद्वितीय हल हैं

Ans. D

Sol. $A^2=AA=\begin{bmatrix} i & i \\ -i & i \end{bmatrix}\begin{bmatrix} i & -i \\ -i & i \end{bmatrix}=\begin{bmatrix} -2 & 2 \\ 2 & -2 \end{bmatrix}$

$A^4=A^2A^2=\begin{bmatrix} -2 & 2 \\ 2 & -2 \end{bmatrix}\begin{bmatrix} -2 & 2 \\ 2 & -2 \end{bmatrix}=\begin{bmatrix} 8 & -8 \\ -8 & 8 \end{bmatrix}$

$A^8=A^4A^4=\begin{bmatrix} 8 & -8 \\ -8 & 8 \end{bmatrix}\begin{bmatrix} 8 & -8 \\ -8 & 8 \end{bmatrix}=\begin{bmatrix} 128 & -128 \\ -128 & 128 \end{bmatrix}$

$128x - 128y = 8$

$-128x + 128y = 64$

system will have no solution

134. If A and B are two non-zero n × n matrics such that $A^2 + B = A^2 B$, then :

यदि A तथा B, n × n के दो शून्येत्तर आव्यूह इस प्रकार हैं कि $A^2 + B = A^2 B$ है, तो :

(A) $A^2B = BA^2$ (B) $A^2 = I$ or $B = I$ (C) $AB = I$ (D) $A^2B = I$

Ans. A

Sol. $A^2 + B = A^2 B$

$(A^2 - I)(B - I)$ .....(1)

$A^2 + B = A^2 B$

$A^2 (B - I) = B$

$A^2 B (B - I)^{-1}$

$A^2 = B (A^2 - I)$

$A^2 = BA^2 - B$

$A^2 + B = BA^2$

$A^2B = BA^2$

135. Let α be a root of the equation $(a-c)x^2 + (b-a)x + (c-b) = 0$ where a, b, c are distinct real numbers such that the matrix $\begin{bmatrix} \alpha^2 & \alpha & 1 \\ 1 & 1 & 1 \\ a & b & c \end{bmatrix}$ is singular. Then, the value of $\frac{(a-c)^2}{(b-a)(c-b)} + \frac{(b-a)^2}{(a-c)(c-b)} + \frac{(c-b)^2}{(a-c)(b-a)}$ is :

माना समीकरण $(a-c)x^2 + (b-a)x + (c-b) = 0$, का मूल α है तथा आव्यूह जहाँ a, b, c भिन्न वास्तविक संख्याऐं है जबकि आव्यूह $\begin{bmatrix} \alpha^2 & \alpha & 1 \\ 1 & 1 & 1 \\ a & b & c \end{bmatrix}$ अव्युत्क्रमणीय है। तो $\frac{(a-c)^2}{(b-a)(c-b)} + \frac{(b-a)^2}{(a-c)(c-b)} + \frac{(c-b)^2}{(a-c)(b-a)}$ का मान है :

(A) 3 (B) 12 (C) 6 (D) 9

Ans. A

Sol. $\Delta = 0 = \begin{vmatrix} \alpha^2 & \alpha & 1 \\ 1 & 1 & 1 \\ a & b & c \end{vmatrix}$

$\Rightarrow \alpha^2(c-b) - \alpha(c-a) + (b-a) = 0$

It is singular when $\alpha = 1$

$$\frac{(a-c)^2}{(b-a)(c-b)} + \frac{(b-a)^2}{(a-c)(c-b)} + \frac{(c-b)^2}{(a-c)(b-a)}$$

$$\frac{(a-b)^3 + (b-c)^3 + (c-a)^3}{(a-b)(b-c)(c-a)}$$

$$= 3\frac{(a-b)(b-c)(c-a)}{(a-b)(b-c)(c-a)} = 3$$

136. The number of square matrices of order 5 with entries from the set {0, 1}, such that the sum of all the elements in each row is 1 and the sum of all the elements in each column is also 1, is :

कोटि 5 के वर्ग आव्यूहों, जिनके अवयव समुच्चय {0, 1} से है, प्रत्येक पंक्ति के सभी अवयवों का योग 1 है तथा प्रत्येक स्तम्भ के सभी

अवयवों का योग भी 1 है, की संख्या है :

(A) 225 (B) 125 (C) 150 (D) 120

Ans. D

Sol.

$$\begin{bmatrix} - & - & - & - & - & - & - & - & - & - & - & - \\ - & - & - & - & - & - & - & - & - & - & - & - \\ - & - & - & - & - & - & - & - & - & - & - & - \\ - & - & - & - & - & - & - & - & - & - & - & - \\ - & - & - & - & - & - & - & - & - & - & - & - \end{bmatrix}$$

In each row and each column exactly one is to be placed :
$\therefore$ No. of such matrices = $5 \times 4 \times 3 \times 2 \times 1 = 120$
Alternate :

$$\begin{bmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{matrix} \rightarrow 5\text{ways} \\ \rightarrow 4\text{ways} \\ \rightarrow 3\text{ways} \\ \rightarrow 4\text{ways} \\ \rightarrow 1\text{ways} \end{matrix}$$

Step-1 : Select any 1 place for 1's in row 1. Automatically some column will get filled with 0's.
Step-2 : From next now select 1 place for 1's. Automatically some column will get filled with 0's.
$\Rightarrow$ Each time one less place will be available for putting 1's.
Repeat step-2 till last row.
Req. ways = $5 \times 4 \times 3 \times 2 \times 1 = 120$

137. Let x, y, z > 1 and $A = \begin{bmatrix} 1 & \log_x y & \log_x z \\ \log_y x & 2 & \log_y z \\ \log_z x & \log_z y & 3 \end{bmatrix}$. Then $|\text{adj}(\text{adj } A^2)|$ is equal to :

माना x, y, z > 1 है तथा $A = \begin{bmatrix} 1 & \log_x y & \log_x z \\ \log_y x & 2 & \log_y z \\ \log_z x & \log_z y & 3 \end{bmatrix}$ है। तो $|\text{adj}(\text{adj } A^2)|$ बराबर है :

(A) 4^8 (B) 2^8 (C) 6^4 (D) 2^4

Ans. B

Sol. $A = \begin{bmatrix} 1 & \log_x y & \log_x z \\ \log_y x & 2 & \log_y z \\ \log_z x & \log_z y & 3 \end{bmatrix}$

$$|A| = \frac{1}{\log x \log y \log z}\begin{vmatrix} \log x & \log y & \log z \\ \log x & 2\log y & \log z \\ \log x & \log y & 3\log z \end{vmatrix}$$

$$|A| = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 3 \end{vmatrix}$$

$|A| = 2$

$\left|\text{adj}\left(\text{adj}\, A^2\right)\right| = |A|^8$

$= 2^8$

138. Let $A = \begin{bmatrix} \frac{1}{\sqrt{10}} & \frac{3}{\sqrt{10}} \\ \frac{-3}{\sqrt{10}} & \frac{1}{\sqrt{10}} \end{bmatrix}$ and $B = \begin{bmatrix} 1 & -i \\ 0 & 1 \end{bmatrix}$, where $i = \sqrt{-1}$

If $M = A^T B A$, then the inverse of the matrix $AM^{2023}A^T$ is

माना $A = \begin{bmatrix} \frac{1}{\sqrt{10}} & \frac{3}{\sqrt{10}} \\ \frac{-3}{\sqrt{10}} & \frac{1}{\sqrt{10}} \end{bmatrix}$ तथा $B = \begin{bmatrix} 1 & -i \\ 0 & 1 \end{bmatrix}$ है, जहाँ $i = \sqrt{-1}$ है।

यदि $M = A^T B A$ है, तो आव्युह $AM^{2023}A^T$ का व्युत्क्रम है

(A) $\begin{bmatrix} 1 & -2023i \\ 0 & 1 \end{bmatrix}$ (B) $\begin{bmatrix} 1 & 0 \\ 2023i & 1 \end{bmatrix}$ (C) $\begin{bmatrix} 1 & 2023i \\ 0 & 1 \end{bmatrix}$ (D) $\begin{bmatrix} 1 & 0 \\ -2023i & 1 \end{bmatrix}$

Ans. C

Sol. $AA^T = \begin{bmatrix} \frac{1}{\sqrt{10}} & \frac{3}{\sqrt{10}} \\ \frac{-3}{\sqrt{10}} & \frac{1}{\sqrt{10}} \end{bmatrix}\begin{bmatrix} \frac{1}{\sqrt{10}} & \frac{-3}{\sqrt{10}} \\ \frac{3}{\sqrt{10}} & \frac{1}{\sqrt{10}} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$B^2 = \begin{bmatrix} 1 & -i \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & -i \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -2i \\ 0 & 1 \end{bmatrix}$

$B^3 = \begin{bmatrix} 1 & -3i \\ 0 & 1 \end{bmatrix}$

.
.
.

$$B^{2023} = \begin{bmatrix} 1 & -2023i \\ 0 & 1 \end{bmatrix}$$

$M = A^T BA$

$M^2 = M \cdot M = A^T BAA^T BA = A^T B^2 A$

$M^3 = M^2 \cdot M = A^T B^2 A A^T BA = A^T B^3 A$

.
.
.

$M^{2023} = \ldots\ldots\ldots\ldots A^T B^{2023} A$

$AM^{2023} A^T = AA^T B^{2023} AA^T = B^{2023}$

$$= \begin{bmatrix} 1 & -2023i \\ 0 & 1 \end{bmatrix}$$

Inverse of $(AM^{2023} A^T)$ is $\begin{bmatrix} 1 & 2023i \\ 0 & 1 \end{bmatrix}$

139. Let A, B, C be 3×3 matrices such that A is symmetric and B and C are skew-symmetric.

Consider the statements

(S1) $A^{13}B^{26} - B^{26}A^{13}$ is symmetric

(S2) $A^{26}C^{13} - C^{13}A^{26}$ is symmetric

Then

(A) Both S1 and S2 are true

(B) Only S1 is true

(C) Both S1 and S2 are false

(D) Only S2 is true

माना 3×3 के आव्यूहों A, B, C में A सममित है तथा B और C विषम सममित है। कथनों

(S1) $A^{13}B^{26} - B^{26}A^{13}$ सममित है

(S2) $A^{26}C^{13} - C^{13}A^{26}$ सममित है

का विचार कीजिए। तो

(A) S1 तथा S2 दोनों सत्य है

(B) केवल S1 सत्य है

(C) S1 तथा S2 दोनों असत्य है

(D) केवल S2 सत्य है

Ans. D

Sol. $S_1 : (A^{13} B^{26} - B^{26} A^{13})^T = (A^{13} B^{26})T - (B^{26} A^{13})^T$

$\Rightarrow (B^{26})^T (A^{13})^T - (A^{13})^T (B^{26})^T$

$\Rightarrow (BT)^{26} (A^T)^{13} - (A^T)^{13} (B^T)^{26}$

$\Rightarrow (-B)^{26} A^{13} - A^{13} (-B)26$

$\Rightarrow -(A^{13}B^{26} - B26\, A^{13})$ so skew symmetric

$S_2 : (A^{26} C^{13} - C^{13} A^{26})^T$

$= (C^T)^{13} (A^T)^{26} - (A^T)^{26} (C^T)^{13}$

$= - C^{13} A^{26} + A^{26} . C^{13}$

So, symmetric matrix

Hence S_2 is true and S_1 is false

140. Let α and β be real numbers. Consider a 3×3 matrix A such that $A^2 = 3A + \alpha I$. If $A^4 = 21A + \beta I$, then

माना α व β वास्तविक संख्याएं है। एक 3×3 आव्यूह A है लिए $A^2 = 3A + \alpha I$. है। यदि $A^4 = 21A + \beta I$ है, तब

(A) $\beta = 8$ (B) $\beta = -8$ (C) $\alpha = 4$ (D) $\alpha = 1$

Ans. B

Sol. $A^2 = 3A + \alpha I$

$A^3 = 3A^2 + \alpha A$

$A^3 = 3(3A + \alpha I) + \alpha A$

$A^3 = 9A + \alpha A + 3\alpha I$

$A^4 = (9 + \alpha) A^2 + 3\alpha A$

$= (9 + \alpha) (3A + \alpha I) + 3\alpha A$

$= A (27 + 6\alpha) + \alpha (9 + \alpha)$

$\Rightarrow 27 + 6\alpha = 21 \Rightarrow \alpha = -1$

$\Rightarrow \beta = \alpha (9 + \alpha) = -8$

141. The set of all values of $t \in R$ for which the matrix

$$\begin{bmatrix} e^t & e^{-1}(\sin t - 2\cos t) & e^{-1}(-2\sin t - \cos t) \\ e^t & e^{-1}(2\sin t + \cos t) & e^{-t}(\sin t - 2\cos t) \\ e^t & e^{-1}\cos t & e^{-t}\sin t \end{bmatrix}$$ is invertible, is

$t \in R$ के सभी मानों, जिनके लिए आव्यूह

$\begin{bmatrix} e^t & e^{-1}(\sin t-2\cos t) & e^{-1}(-2\sin t-\cos t) \\ e^t & e^{-1}(2\sin t+\cos t) & e^{-t}(\sin t-2\cos t) \\ e^t & e^{-1}\cos t & e^{-t}\sin t \end{bmatrix}$ व्युत्क्रमणीय है, का समुच्चय है :

(A) R (B) $\{k\pi, k \in Z\}$ (C) $\left\{(2k+1)\frac{\pi}{2}, k \in Z\right\}$ (D) $\left\{k\pi+\frac{\pi}{4}, k \in Z\right\}$

Ans. A

Sol. $|A| = \begin{bmatrix} e^t & e^{-t}(\sin t-2\cos t) & e^{-t}(-2\sin t-\cos t) \\ e^t & e^{-t}(2\sin t+\cos t) & e^{-t}(\sin t-2\cos t) \\ e^t & e^{-t}\cos t & e^{-t}\sin t \end{bmatrix}$

$= e^t.e^{-t}.e^{-t}\begin{vmatrix} 1 & \sin t-2\cos t & -2\sin t-\cos t \\ 1 & 2\sin t+\cos t & \sin t-2\cos t \\ 1 & \cos t & \sin t \end{vmatrix}$

$R_1 \to R_1 - R_3, R_2 \to R_2 - R_3$

$|A| = e^{-t}\begin{vmatrix} 0 & \sin t-3\cos t & -3\sin t-\cos t \\ 0 & 2\sin t & -2\cos t \\ 1 & \cos t & \sin t \end{vmatrix}$

$= e^{-t}(-2\cos t(\sin t - 3\cos t) + 2\sin t(3\sin t + \cos t))$
$= e^{-t}(6\cos^2 t + 6\sin^2 t) = 6e^{-t}$
$|A| \neq 0 \ \forall \ t \in R$

142. Let $A = \begin{pmatrix} m & n \\ p & q \end{pmatrix}$, $d = |A| \neq 0$ and $|A - d(\text{Adj } A)| = 0$. Then

माना $A = \begin{pmatrix} m & n \\ p & q \end{pmatrix}$, $d = |A| \neq 0$ तथा $|A - d(\text{Adj } A)| = 0$ है। तो :

(A) $1 + d^2 = (m + q)^2$ (B) $(1 + d)^2 = (m + q)^2$

(C) $(1 + d)^2 = m^2 + q^2$ (D) $1 + d^2 = m^2 + q^2$

Ans. B

Sol. $A = \begin{bmatrix} m & n \\ p & q \end{bmatrix}$

$d = mq - np$

$\text{adj}A = \begin{bmatrix} q & -n \\ -p & m \end{bmatrix}$

$|A - d\,\text{Adj}\,A| = 0$

$\begin{bmatrix} m - dq & n + dn \\ p + dp & q - dm \end{bmatrix}$

$(m - dq)(q - dm) - (d + 1)^2 np = 0$

$mq - dm^2 - dq^2 + d^2 mq = (d + 1)^2 np$

$mq(1 + d^2) - d(m^2 + q^2) = (d + 1)^2 (mq - d)$

$mq(1 + d^2) - d(m^2 + q^2) = (d^2 + 2d + 1)\, mq - (d + 1)^2 d$

$(m + q)^2 = (d + 1)^2$

143. If P is 3×3 real matrix such that $P^T = aP + (a - 1)I$, where a > 1, then

(A) $|\text{Adj P}| > 1$ (B) $|\text{Adj P}| = \frac{1}{2}$

(C) $|\text{Adj P}| = 1$ (D) P is a singular matrix

यदि P एक 3×3 का वास्तविक आव्यूह इस प्रकार है कि $P^T = aP + (a - 1)I$ है, जहाँ a > 1 है, तब :

(A) $|\text{Adj P}| > 1$ (B) $|\text{Adj P}| = \frac{1}{2}$

(C) $|\text{Adj P}| = 1$ (D) P एक singular matrix है

Ans. C

Sol. $(P^T)^T = (aP + (a - 1)I)^T$

$\Rightarrow P = aP^T + (a - 1)I$

$\Rightarrow P = a(ap + (a - 1)I) + (a - 1)I$

$\Rightarrow P = a^2 P + a(a - 1)I + (a - 1)I$

$\Rightarrow P = a^2 P + (a^2 - 1)I$

$\Rightarrow (a^2 - 1)P + (a^2 - 1)I = 0$

$\Rightarrow (a^2 - 1)(P + I) = 0$

$\therefore a > 1$

$\Rightarrow P + I = 0$

$\Rightarrow P = -I$

$\text{adj}\,P = \text{adj}\,(-I)$

$\Rightarrow |\text{adj}\,P| = |\text{adj}\,(-I)| = |-I|^2$

$\Rightarrow |\text{adj}\,P| = 1$

144. Let $A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 4 & -1 \\ 0 & 12 & -3 \end{pmatrix}$. Then the sum of the diagonal elements of the matrix $(A + I)^{11}$ is equal to :

माना $A=\begin{pmatrix}1 & 0 & 0\\ 0 & 4 & -1\\ 0 & 12 & -3\end{pmatrix}$ है। तो आव्यूह $(A+I)^{11}$ के विकर्ण के अवयवों का योग है :

(A) 6144 (B) 4094 (C) 2050 (D) 4097

Ans. D

Sol. $A^2=\begin{bmatrix}1 & 0 & 0\\ 0 & 4 & -1\\ 0 & 12 & -3\end{bmatrix}\begin{bmatrix}1 & 0 & 0\\ 0 & 4 & -1\\ 0 & 12 & -3\end{bmatrix}=\begin{bmatrix}1 & 0 & 0\\ 0 & 4 & -1\\ 0 & 12 & -3\end{bmatrix}=A$

Now $(A+I)^{11}={}^{11}C_0\,A^{11}+{}^{11}C_1+{}^{11}C_1\,A^{10}+\ldots\ldots+{}^{11}C_{11}\,I$

$=A\,({}^{11}C_0+{}^{11}C_1+\ldots\ldots+{}^{11}C_{10})+I$

$=A\,(2^{11}-1)+I$

Trace of $(A+I)^{11}$

$=2^{11}+4(2^{11}-1)+1+(-3)\,(2^{11}-1)+1$

$=2\times 2^{11}+1$

$=2^{12}+1$

145. Let $A=\left[a_{ij}\right]_{2\times 2}$, where $a_{ij}\neq 0$ for all i, j and $A^2=I$. Let a be the sum of all diagonal elements of A and $b=|A|$. Then $3a^2+4b^2$ is equal to :

माना $A=\left[a_{ij}\right]_{2\times 2}$, जहाँ सभी i, j के लिए $a_{ij}\neq 0$ एवं $A^2=I$ है। माना A के विकर्ण के सभी अवयवों का योग a है और $b=|A|$ है। तब $3a^2+4b^2$ बराबर है :

(A) 7 (B) 3 (C) 14 (D) 4

Ans. D

Sol. Let $A=\begin{bmatrix}p & q\\ r & s\end{bmatrix}$

$A^2=\begin{bmatrix}p & q\\ r & s\end{bmatrix}\begin{bmatrix}p & q\\ r & s\end{bmatrix}=\begin{bmatrix}p^2+qr & pq+qs\\ rp+rs & qr+s^2\end{bmatrix}$

$A^2=I$

$\Rightarrow p^2+qr=1 \quad q(p+s)=0$

$r(p+s)=0 \qquad qr+s^2=1$

$q\neq 0\Rightarrow p+s=0\Rightarrow a=0$

$b=|A|=ps-qr=-p^2-qr=-1(\because s=-p)$

$\therefore 3a^2+4b^2=4$

146. Let P be a square matrix such that $P^2=I-P$. For $\alpha,\beta,\gamma,\delta\in\mathbb{N}$, if $P^{\alpha}+P^{\beta}=\gamma I-29P$ and $P^{\alpha}-P^{\beta}=\delta I-13P$,

then $\alpha+\beta+\gamma-\delta$ is equal to :

माना P एक वर्ग आव्यूह है जिसके लिए $P^2 = I - P$ है। तो $\alpha,\beta,\gamma,\delta \in \mathbb{N}$ के लिए यदि $P^\alpha + P^\beta = \gamma I - 29P$ तथा $P^\alpha - P^\beta = \delta I - 13P$ है, तो $\alpha+\beta+\gamma-\delta$ बराबर है :

(A) 18 (B) 22 (C) 40 (D) 24

Ans. D

Sol. $P^2 = I - P$

$P^\alpha + P^\beta = \gamma I - 29P$

$P^\alpha - P^\beta = \delta I - 13P$

$P^4 = (I-P)^2 = I + P^2 - 2P$

$P^4 = I + I - P - 2P = 2I - 3P$

$P^8 = \left(P^4\right)^2 = (2I-3P)^2 = 4I + 9P^2 - 12P$

$= 4I + 9(I-P) - 12P$

$P^8 = 13I - 21P$

$P^6 = P^4 \cdot P^2 \quad = (2I-3P)(I-P)$

$= 2I - 5P + 3P^2$

$= 2I - 5P + 3(I-P)$

$= 5I - 8P$

$(1)+(2)$ $(1)-(2)$

$P^8 + P^6 = 18I - 29P$ $P^8 - P^6 = 8I - 13P$

From (A) $\alpha = 8, \quad \beta = 6$

$\gamma = 18$

$\delta = 8$

$\alpha+\beta+\gamma-\delta = 32 - 8 = 24$

147. If $A = \dfrac{1}{5!6!7!}\begin{bmatrix} 5! & 6! & 7! \\ 6! & 7! & 8! \\ 7! & 8! & 9! \end{bmatrix}$, then $|\text{adj}(\text{adj}(2A))|$ is equal to :

यदि $A = \dfrac{1}{5!6!7!}\begin{bmatrix} 5! & 6! & 7! \\ 6! & 7! & 8! \\ 7! & 8! & 9! \end{bmatrix}$ है, तो $|\text{adj}(\text{adj}(2A))|$ बराबर है :

(A) 2^{16} (B) 2^8 (C) 2^{20} (D) 2^{12}

Ans. A

Sol. $|A|=\frac{1}{5!6!7!}\times 5!6!7!\begin{vmatrix}1 & 6 & 42\\ 1 & 7 & 56\\ 1 & 8 & 72\end{vmatrix}$

$R_2 \to R_2 - R_1$ & $R_3 \to R_3 - R_2$

$|A|=\begin{vmatrix}1 & 6 & 42\\ 0 & 1 & 14\\ 0 & 1 & 16\end{vmatrix} \Rightarrow |A|=2$

$|\text{Adj}\left(\text{adj}(2A)\right)|=|2A|^{(n-1)^2}=|2A|^4=2^{12}|A|^4=2^{16}$

148. Let A be a 2×2 matrix with real entries such that $A' = \alpha A + I$, where $\alpha \in \mathbb{R}-\{-1,1\}$. If $\det\left(A^2-A\right)=4$, then the sum of all possible values of α is equal to :

माना A वास्तविक अवयवों का एक 2×2 आव्यूह है जिसके लिए $A' = \alpha A + I$ है, $\alpha \in \mathbb{R}-\{-1,1\}$ है। यदि $\det\left(A^2-A\right)=4$ है, तो α के सभी संभव मानों का योग बराबर है :

(A) $\frac{3}{2}$ (B) 2 (C) $\frac{5}{2}$ (D) 0

Ans. C

Sol. $A^T = \alpha A + I$

$A = \alpha A^T + I$

$A = \alpha(\alpha A + I) + I$

$A = \alpha^2 A + (\alpha+1)I$

$A\left(1-\alpha^2\right) = (\alpha+1)I$

$A = \frac{I}{1-\alpha}\ldots(1)$

$|A| = \frac{1}{(1-\alpha)^2}\ldots(2)$

$\left|A^2-A\right| = |A||A-1|\ldots(3)$

$A - I = \frac{I}{I-\alpha} - I = \frac{\alpha}{1-\alpha}I$

$|A-I| = \left(\frac{\alpha}{1-\alpha}\right)^2\ldots(4)$

Now $\left|A^2-A\right| = 4$

$|A||A-I|=4$

$\Rightarrow \frac{1}{(1-\alpha)^2}\frac{\alpha^2}{(1-\alpha^2)}=4$

$\Rightarrow \frac{\alpha}{(1-\alpha)^2}=\pm 2$

$\Rightarrow 2(1-\alpha)^2=\pm\alpha$

$(C_1)\ 2(1-\alpha)^2=\alpha$ $\qquad$ $(C_2)\ 2(1-\alpha)^3=-\alpha$

$2\alpha^2-5\alpha+2=0 \Rightarrow \alpha_1, \alpha_2$ $\qquad$ $2\alpha^2-3\alpha+2=0$

$\alpha_1+\alpha_2=\frac{5}{2}$ $\qquad$ $\alpha \notin R$

149. Let $S=\left\{M=\left[a_{ij}\right], a_{ij}\in\{0,1,2\}, 1\le i,j\le 2\right\}$ be a sample space and $A=\{M\in S: M$ is invertible$\}$ be an event. Then P(A) is equal to :

माना $S=\left\{M=\left[a_{ij}\right], a_{ij}\in\{0,1,2\}, 1\le i,j\le 2\right\}$ एक प्रतिदर्श समष्टि है तथा $A=\{M\in S: M$ व्युत्क्रमणीय है$\}$ एक घटना है। तो P(A) बराबर है :

(A) $\frac{50}{81}$ (B) $\frac{47}{81}$ (C) $\frac{49}{81}$ (D) $\frac{16}{27}$

Ans. A

Sol. $M\begin{bmatrix} a & b \\ c & d \end{bmatrix}$, where $a,b,c,d,\in\{0,1,2\}$

$n(s)=3^4=81$

we first find $p(\bar{A})$

$|m|=0\Rightarrow ad=bc$

$ad=bc=0\Rightarrow$ no$\cdot$ of $(a,b,c,d)=\left(3^2-2^2\right)^2=25$

$ad=bc=1\Rightarrow$ no. of $(a,b,c,d)=1^2=1$

$ad=bc=2\Rightarrow$ no$\cdot$ of $(a,b,c,d)=2^2=4$

$ad=bc=4\Rightarrow$ no. of $(a,b,c,d)=1^2=1$

$\therefore P(\bar{A})=\frac{31}{81}\Rightarrow p(A)=\frac{50}{81}$

150. Let $A=\begin{bmatrix}1 & \frac{1}{51}\\ 0 & 1\end{bmatrix}$. If $B=\begin{bmatrix}1 & 2\\ -1 & -1\end{bmatrix}A\begin{bmatrix}-1 & -2\\ 1 & 1\end{bmatrix}$, then the sum of all the elements of the matrix $\sum_{n=1}^{50}B^n$ is equal to :

माना $A=\begin{bmatrix}1 & \frac{1}{51}\\ 0 & 1\end{bmatrix}$ है। यदि $B=\begin{bmatrix}1 & 2\\ -1 & -1\end{bmatrix}A\begin{bmatrix}-1 & -2\\ 1 & 1\end{bmatrix}$ है, तो $\sum_{n=1}^{50}B^n$ के सभी अवयवों का योग है :

(A) 75 (B) 100 (C) 50 (D) 125

Ans. B

Sol. $\because \overset{(M)}{\begin{bmatrix}1 & 2\\ -1 & -1\end{bmatrix}}\overset{(N)}{\begin{bmatrix}-1 & -2\\ 1 & 1\end{bmatrix}}=\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$

$\therefore \quad MN=I=NM$

$B=MAN$

$B^n=(MAN)^n=(MAN)(MAN)\ldots$

$=\left(MA^2N\right)MAN$

$=MA^nN$

$\text{Now } A=\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}+\begin{bmatrix}0 & \frac{1}{51}\\ 0 & 0\end{bmatrix}=I+E$

$E^2=0$

$\therefore A^n=(I+E)^n=I+nE+\underbrace{{}^nC_2E^2}_{0}$

$=I+nE$

$=\begin{bmatrix}1 & \frac{n}{51}\\ 0 & 1\end{bmatrix}$

$\text{So, } B^n=MA^nN=\begin{bmatrix}1+\frac{n}{51} & \frac{n}{51}\\ \frac{-n}{51} & 1-\frac{n}{51}\end{bmatrix}$

$$\sum_{n=1}^{50} B^n = \begin{bmatrix} 50+\frac{50\cdot 51}{2\cdot 51} & \frac{50\cdot 51}{2\cdot 51} \\ \frac{-50\cdot 51}{2\cdot 51} & 50-\frac{50\cdot 51}{2\cdot 51} \end{bmatrix} = \begin{bmatrix} 75 & 25 \\ -25 & 25 \end{bmatrix}$$

$\therefore$ sum = 100

151. The number of symmetric matrices of order 3, with all the entries from the set {0, 1, 2, 3, 4, 5, 6, 7, 8, 9}, is:

कोटि 3 के सममित आव्यूहों, जिनके सभी अवयव समुच्चय {0, 1, 2, 3, 4, 5, 6, 7, 8, 9} से हैं, की संख्या है :

(A) 6^{10} (B) 10^6 (C) 9^{10} (D) 10^9

Ans. B

Sol. $A = \begin{bmatrix} a & b & c \\ b & d & e \\ c & e & f \end{bmatrix}, a,b,c,d,e,f \in \{0,1,2,.....9\}$

Number of matrices = 10^6

152. Let $B = \begin{bmatrix} 1 & 3 & \alpha \\ 1 & 2 & 3 \\ \alpha & \alpha & 4 \end{bmatrix}$, $\alpha > 2$ be the adjoint of a matrix A and $|A| = 2$. Then $\begin{bmatrix} \alpha - 2\alpha & \alpha \end{bmatrix} B \begin{bmatrix} \alpha \\ -2\alpha \\ \alpha \end{bmatrix}$ is equal to :

माना $B = \begin{bmatrix} 1 & 3 & \alpha \\ 1 & 2 & 3 \\ \alpha & \alpha & 4 \end{bmatrix}$, $\alpha > 2$, एक आव्यूह A का सहखण्डज है तथा $|A| = 2$ है। तो $\begin{bmatrix} \alpha - 2\alpha & \alpha \end{bmatrix} B \begin{bmatrix} \alpha \\ -2\alpha \\ \alpha \end{bmatrix}$ बराबर है :

(A) –16 (B) 16 (C) 0 (D) 32

Ans. A

Sol. $|B| = |\text{adj } A| = |A|^{3-1} = |A|^2 = 4$

$1(8 - 3\alpha) - 3(4 - 3\alpha) + \alpha(\alpha - 2\alpha) = 4$

$\Rightarrow 8 - 3\alpha - 12 + 9\alpha - \alpha^2 = 4$

$\Rightarrow \alpha^2 - 6\alpha + 8 = 0$

$\Rightarrow \alpha = 2, 4$

$\Rightarrow \alpha = 4 \qquad (\because \alpha > 2)$

Now

$$\begin{bmatrix} 4 & -8 & 4 \end{bmatrix} \begin{bmatrix} 1 & 3 & 4 \\ 1 & 2 & 3 \\ 4 & 4 & 4 \end{bmatrix} \begin{bmatrix} 4 \\ -8 \\ 4 \end{bmatrix}$$

$$= 16\begin{bmatrix}1 & -2 & 1\end{bmatrix}\begin{bmatrix}1 & 3 & 4\\ 1 & 2 & 3\\ 4 & 4 & 4\end{bmatrix}\begin{bmatrix}1\\ -2\\ 1\end{bmatrix}$$

$$= 16\begin{bmatrix}1-2+4 & 3-4+4 & 4-6+4\end{bmatrix}\begin{bmatrix}1\\ -2\\ 1\end{bmatrix}$$

$= 16\,[3 - 6 + 2]$

$= 16\,[-1] = [-16]$

153. Let the determinant of a square matrix A of order m be m – n, where m and n satisfy 4m + n = 22 and 17m + 4n = 93. If $\det(n\,\text{adj}(\text{adj}(mA))) = 3^a5^b6^c$, then a + b + c is equal to :

यदि m कोटि के वर्ग आव्यूह A के सारणिक का मान m – n है, जहाँ m तथा n, 4m + n = 22 तथा 17m + 4n = 93 को संतुष्ट करते है। यदि $\det(n\,\text{adj}(\text{adj}(mA))) = 3^a5^b6^c$ हो, तो a + b + c के मान बराबर होगा :

(A) 96 (B) 101 (C) 109 (D) 84

Ans. A

Sol. $|A| = m - n$

$4m + n = 22$

$17m + 4n = 93$

$m = 5, n = 2$

$|A| = 3$

$|2\ \text{adj}\ (\text{adj}\ 5A)| = 2^5\,|5A|^{16}$

$= 2^5 \cdot 5^{80}\,|A|^{16}$

$= 2^5 \cdot 5^{80} \cdot 3^{16}$

$= 3^{11} \cdot 5^{80} \cdot 6^5$

$a + b + c = 96$

154. Let $P = \begin{bmatrix}3 & -1 & -2\\ 2 & 0 & \alpha\\ 3 & -5 & 0\end{bmatrix}$, where $\alpha \in R$. Suppose $Q = [q_{ij}]$ is a matrix satisfying $PQ = kI_3$ for some non–zero $k \in R$.

If $q_{23} = -\frac{k}{8}$ and $|Q| = \frac{k^2}{2}$ then $\alpha^2 + k^2$ is equal to _____ .

माना $P = \begin{bmatrix}3 & -1 & -2\\ 2 & 0 & \alpha\\ 3 & -5 & 0\end{bmatrix}$ है, जबकि $\alpha \in R$ है। माना $Q = [q_{ij}]$ एक आव्यूह है, जिसके लिए $PQ = kI_3$, किसी शून्येतर, $k \in R$

के लिए, है। यदि $q_{23} = -\frac{k}{8}$ तथा $|Q| = \frac{k^2}{2}$ हैं, तो $\alpha^2 + k^2$ बराबर है _____ ।

Ans. 17

Sol. $P = \begin{bmatrix} 3 & -1 & -2 \\ 2 & 0 & \alpha \\ 3 & -5 & 0 \end{bmatrix}$

$|P| = 4(3\alpha + 5)$

$PQ = kI_3$

$|P| \cdot |Q| = k^3$

$4(3\alpha + 5) \cdot \frac{k^2}{2} = k^3$

$k = 2(3\alpha + 5) \ldots\ldots(1)$

$Q = k P^{-1}$

$Q = \frac{k \, adj.P}{|P|}$

$q_{23} = \frac{k}{4(3\alpha + 5)} . C_{32}$ $(C_{ij} = \text{Cofacor of } a_{ij})$

$\frac{-k}{8} = \frac{k}{4(3\alpha + 5)} \times (-1) \times (3\alpha + 6)$

$\alpha = -1$

$\alpha^2 + k^2 = 17$

155. Let $A = \begin{bmatrix} x & y & z \\ y & z & x \\ z & x & y \end{bmatrix}$, where x, y and z are real numbers such that $x + y + z > 0$ and $xyz = 2$. If $A^2 = I_3$, then the value of $x^3 + y^3 + z^3$ is _____.

माना $A = \begin{bmatrix} x & y & z \\ y & z & x \\ z & x & y \end{bmatrix}$ है, जहाँ x, y तथा z वास्तविक संख्याऐं हैं, जिनके लिए $x + y + z > 0$ तथा $xyz = 2$ हैं।

यदि $A^2 = I_3$ है, तो $x^3 + y^3 + z^3$ का मान है :

Ans. 7

Sol. $\begin{bmatrix} x & y & z \\ y & z & x \\ z & x & y \end{bmatrix}\begin{bmatrix} x & y & z \\ y & z & x \\ z & x & y \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$$\begin{bmatrix} x^2+y^2+z^2 & xy+yz+zx & xy+yz+zx \\ xy+yz+zx & x^2+y^2+z^2 & xy+yz+zx \\ xy+yz+zx & xy+yz+zx & x^2+y^2+z^2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$x^2 + y^2 + z^2 = 1$

$xy + yz + zx = 0$

$(x + y + z)^2 = \sum x^2 + 2\sum xy$

$(x + y + z)^2 = 1 + 0$

$x + y + z = \pm 1$

$\Rightarrow x + y + z = 1$

$x^3 + y^3 + z^3 = x^3 + y^3 + z^3 - 3xyz + 3xyz$

$= (x + y + z)(x^2 + y^2 + z^2 - xy - yz - zx) + 3xyz$

$= (1)(1) + 3 \times 2$

$= 7$

156. If the matrix $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 3 & 0 & -1 \end{bmatrix}$ satisfies the equation $A^{20} + \alpha A^{19} + \beta A \begin{bmatrix} 1 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ for some real numbers α and β, then $\beta - \alpha$ is equal to _____.

यदि किसी वास्तविक संख्याओं α तथा β के लिए आव्यूह $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 3 & 0 & -1 \end{bmatrix}$, समीकरण

$A^{20} + \alpha A^{19} + \beta A \begin{bmatrix} 1 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ को सन्तुष्ट करता है, तो $\beta - \alpha$ बराबर है _____।

Ans. 4

Sol. $A=\begin{bmatrix}1 & 0 & 0\\ 0 & 2 & 0\\ 3 & 0 & -1\end{bmatrix}$

$$A^2=\begin{bmatrix}1 & 0 & 0\\ 0 & 3 & 0\\ 3 & 0 & -1\end{bmatrix}\begin{bmatrix}1 & 0 & 0\\ 0 & 2 & 0\\ 3 & 0 & -1\end{bmatrix}$$

$$A^2=\begin{bmatrix}1 & 0 & 0\\ 0 & 2^2 & 0\\ 0 & 0 & 1\end{bmatrix}$$

$$A^3=\begin{bmatrix}1 & 0 & 0\\ 0 & 2^3 & 0\\ 3 & 0 & -1\end{bmatrix}$$

$$A^4=\begin{bmatrix}1 & 0 & 0\\ 0 & 2^4 & 0\\ 0 & 0 & 1\end{bmatrix}$$

By observing the pattren

$$A^{20}=\begin{bmatrix}1 & 0 & 0\\ 0 & 2^{20} & 0\\ 0 & 0 & 1\end{bmatrix}$$

$$A^{19}=\begin{bmatrix}1 & 0 & 0\\ 0 & 2^{19} & 0\\ 3 & 0 & -1\end{bmatrix}$$

$$A^{20}+2A^{19}+\beta A=\begin{bmatrix}1+\alpha+\beta & 0 & 0\\ 0 & 2^{20}+\alpha\cdot 2^{19}+2\beta & 0\\ 3\alpha+3\beta & 0 & 1-\alpha-\beta\end{bmatrix}$$

$\alpha + \beta = 0$ and $2^{20} + \alpha \cdot 2^{19} + 2\beta = 4$

$2^{20} + \alpha \cdot 2^{19} - 2\alpha = 4$

$\Rightarrow 2^{19}(\alpha + 2) = 2(\alpha + 2)$

$(\alpha + 2)(2^{19} - 2) = 0$

$\alpha = -2$

$\therefore \beta = 2$

$\therefore |\alpha - \beta| = |-2 -2| = |4|$

$\beta - \alpha = 2 - (-2) = 4.$

157. If $A = \begin{bmatrix} 2 & 3 \\ 0 & -1 \end{bmatrix}$, then the value of $\det(A^4) + \det(A^{10} - (\text{Adj}(2A))^{10})$ is equal to :

यदि $A = \begin{bmatrix} 2 & 3 \\ 0 & -1 \end{bmatrix}$ है, तो $\det(A^4) + \det(A^{10} - (\text{Adj}(2A))^{10})$ का मान बराबर है :

Ans. 16

Sol. $|A| = -2$

$|A|^4 = 16$

$A^{10} = \begin{bmatrix} 2^{10} & 2^{10}-1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1024 & 1023 \\ 0 & 1 \end{bmatrix}$

$2A = \begin{bmatrix} 4 & 6 \\ 0 & -2 \end{bmatrix}$

$\text{adj}\,(2A) = \begin{bmatrix} -2 & -6 \\ 0 & 4 \end{bmatrix}$

$(\text{adj}\,(2A))^{10} = 2^{10} \begin{bmatrix} 1 & 3 \\ 0 & -2 \end{bmatrix}^{10}$

$= 2^{10} \begin{bmatrix} 1 & -\left(2^{10}-1\right) \\ 0 & 2^{10} \end{bmatrix}$

$A^{10} - (\text{adj}\,(2A))^{10} = \begin{bmatrix} 0 & 2^{11} \times 1023 \\ 0 & 1-(1024)^2 \end{bmatrix}$

$\left|A^{10} - \left(\text{adj}(2a)\right)^{10}\right| = 0$

158. Let $A = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}$ and $B = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$ be two 2×1 matrices with real entries such that $A = XB$, where $X = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & -1 \\ 1 & k \end{bmatrix}$ and $k \in R$. If $a_1^2 + a_2^2 = \frac{2}{3}\left(b_1^2 + b_2^2\right)$ and $\left(k^2+1\right)b_2^2 \neq -2b_1b_2$ then the value of k is _____.

माना 2×1 के दो आव्यूह $A=\begin{bmatrix} a_1 \\ a_2 \end{bmatrix}$ तथा $B=\begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$ हैं जिनके अव्यव वास्तविक हैं तथा $A = XB$ है, जहाँ $X=\frac{1}{\sqrt{3}}\begin{bmatrix} 1 & -1 \\ 1 & k \end{bmatrix}$ और $k \in R$ है। यदि $a_1^2 + a_2^2 = \frac{2}{3}\left(b_1^2 + b_2^2\right)$ तथा $\left(k^2+1\right)b_2^2 \neq -2b_1b_2$ हैं, तो k का मान है _____।

Ans. 1

Sol. $A = XB$

$$\begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = \frac{1}{\sqrt{3}}\begin{bmatrix} 1 & -1 \\ 1 & k \end{bmatrix}\begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$$

$$a_1 = \frac{1}{\sqrt{3}}\left(b_1 - b_2\right)$$

$$a_2 = \frac{1}{\sqrt{3}}\left(b_1 + kb_2\right)$$

$$a_1^2 + a_2^2 = \frac{1}{3}\left\{\left(b_1 - b_2\right)^2 + \left(b_1 + kb_2\right)^2\right\}$$

$$\frac{2}{3}\left(b_1^2 + b_2^2\right) = \frac{1}{3}\left\{\left(b_1 - b_2\right)^2 + \left(b_1 + kb_2\right)^2\right\}$$

$$2\left(b_1^2 + b_2^2\right) = \left\{\left(b_1 - b_2\right)^2 + \left(b_1 + kb_2\right)^2\right\}$$

$k = 1$

159. Let $P=\begin{bmatrix} -30 & 20 & 56 \\ 90 & 140 & 112 \\ 120 & 60 & 14 \end{bmatrix}$ and $A=\begin{bmatrix} 2 & 7 & \omega^2 \\ -1 & -\omega & 1 \\ 0 & -\omega & \omega+1 \end{bmatrix}$ where $\omega=\frac{-1+i\sqrt{3}}{2}$, and I_3 be the identity matrix of order 3. If the determinant of the matrix $(P^{-1}AP - I_3)^2$ is $\alpha\omega^2$, then the value of α is equal to _____.

माना $P=\begin{bmatrix} -30 & 20 & 56 \\ 90 & 140 & 112 \\ 120 & 60 & 14 \end{bmatrix}$ तथा $A=\begin{bmatrix} 2 & 7 & \omega^2 \\ -1 & -\omega & 1 \\ 0 & -\omega & \omega+1 \end{bmatrix}$ है, जहां $\omega=\frac{-1+i\sqrt{3}}{2}$ है तथा 3 कोटी का तत्समक आव्यूह I_3 है। यदि आव्यूह $(P^{-1}AP - I_3)^2$ का सारणिक $\alpha\omega^2$ है, तो α का मान बराबर है _____ ।

Ans. 36

Sol. $M = (P^{-1}AP - I)(P^{-1}AP - I)$

$= P^{-1}AP\,P^{-1}AP - P^{-1}AP - P^{-1}AP + I$

$= P^{-1}A^2P - 2P^{-1}AP + P^{-1}IP$

$= P^{-1}(A^2 - 2A + I)P$

$M = P^{-1}(A - I)^2 P$

$|M| = |P^{-1}|\,|A - I|^2\,|P|$

$= |A - I|^2$

$$|A - I|^2 = \begin{vmatrix} 1 & 7 & \omega^2 \\ -1 & -1-\omega & 1 \\ 0 & -\omega & -\omega \end{vmatrix}^2$$

$$= \begin{vmatrix} 1 & 7 & \omega^2 \\ -1 & \omega^2 & 1 \\ 0 & -\omega & -\omega \end{vmatrix}^2$$

$= \left(-1+\omega+7(-\omega)+\omega^2(w)\right)^2$

$= (-6\omega)^2 = 36\omega^2$

$\Rightarrow \alpha = 36$

160. The total number of 3×3 matrices A having entries from the set $\{0, 1, 2, 3\}$ such that the sum of all the diagonal entries of AA^T is 9, is equal to _____.

3×3 के आव्यूहों A, जिनके अवयव समुच्चय $\{0, 1, 2, 3\}$ में से हैं तथा AA^T के विकर्ण के सभी अवयवों का योगफल 9 है, की कुल संख्या है _____ ।

Ans. 766

Sol. $$A = \begin{bmatrix} a_a & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix}$$

Sum of diagonal entries of $AA^T = \sum a_i^2 + \sum b_i^2 + \sum c_1^2$

$\{0, 0, 0, ...0, 3\} \Rightarrow \frac{\lfloor 9}{\lfloor 8} = 9$

$\{0, 0, 0, 1, 1, 1, 1, 1, 2\}\ \frac{\lfloor 9}{\lfloor 3 \lfloor 5 \lfloor 1} = 504$

$\{2, 2, 1, 0, 0, 0, 0, 0, 0\}\ \frac{\lfloor 9}{\lfloor 2 \lfloor 6} = 252$

$\{1, 1, 1, 1, 1, 1, 1, 1, 1\}\ \frac{\lfloor 9}{\lfloor 9} = 1$

Total = 9 + 504 + 252 + 1 = 766

161. Let A be the symmetric matrix such that $|A| = 2$ and $\begin{bmatrix} 2 & 1 \\ 3 & \frac{3}{2} \end{bmatrix} A = \begin{bmatrix} 1 & 2 \\ \alpha & \beta \end{bmatrix}$. If the sum of the diagonal elements of A is s, then $\frac{\beta s}{\alpha^2}$ is equal to _____ .

माना A एक सममित आव्यूह है जिसकी सारणिक 2 है तथा $\begin{bmatrix} 2 & 1 \\ 3 & \frac{3}{2} \end{bmatrix} A = \begin{bmatrix} 1 & 2 \\ \alpha & \beta \end{bmatrix}$ है। यदि A के विकर्ण के अवयवों का योग s है, तो $\frac{\beta s}{\alpha^2}$ बराबर है।

Ans. 5

Sol. $A^T = A \Rightarrow$ So, $A = \begin{bmatrix} a & b \\ b & d \end{bmatrix}$

now $ad - b^2 = 2$ ...(1)

also $BA = \begin{bmatrix} 2 & 1 \\ 3 & 3/2 \end{bmatrix} \begin{bmatrix} a & b \\ b & d \end{bmatrix}$

$\Rightarrow 2a + b = 1$ ...(2)

$3a + \frac{3b}{2} = \alpha$ ...(3)

$3b + \frac{3d}{2} = \beta$ ...(4)

$2b + d = 2$ ...(5)

Solving equations

$a = \frac{3}{4}, \ \alpha = \frac{3}{2}, \ b = -\frac{1}{2}$

$d = 3, \ \beta = 3$

So $\frac{\beta \times s}{\alpha} = 5$

162. Let A be a $n \times n$ matrix such that $|A| = 2$. If the determinant of the matrix $\text{Adj}\,(2 \cdot \text{Adj}\,(2A^{-1})) \cdot$ is 2^{84}, then n is equal to _____ .

माना A एक $n \times n$ आव्यूह है तथा $|A| = 2$ है। यदि आव्यूह $\text{Adj}\,(2 \cdot \text{Adj}\,(2A^{-1}))$ का सारणिक 2^{84} है, तो n बराबर है।

Ans. 5

Sol. $|A| = 2$

Hence Adj (2. Adj $\left(2A^{-1}\right)\Big) = \left|2.\text{Adj}\left(2A^{-1}\right)\right|^{n-1}$

$= \left(2^n \left|\text{Adj}\left(2A^{-1}\right)\right|\right)^{n-1}$

$= \left(2^n \left|2A^{-1}\right|^{n-1}\right)^{n-1}$

$= 2^{n(n-1)}\left(\left(2^n \left|A^{-1}\right|\right)^{(n-1)}\right)^{(n-1)} \quad \therefore \left|A^{-1}\right| = \frac{1}{|A|} = \frac{1}{2}$

$= 2^{n(n-1)}\left(\left(2^{(n-1)}\right)^{(n-1)}\right)^{(n-1)}$

$= 2^{n(n-1)+(n-1)^3} = 2^{84}$

$\Rightarrow n = 5$

163. Let $A = [a_{ij}], a_{ij} \in Z \cap [0,4], 1 \le i, j \le 2$. The number of matrices A such that the sum of all entries is a prime number $p \in (2,13)$ is _____ .

माना $A = [a_{ij}], a_{ij} \in Z \cap [0,4], 1 \le i, j \le 2$ है। ऐसे आव्यूहों A जिनके सभी अवयवों का योग एक अभाज्य संख्या $p \in (2,13)$ है, की संख्या _____ है।

Ans. 196

Sol. As given a + b + c + d = 3 or 5 or 7 or 11

if sum = 3

$\left(1 + x + x^2 + \ldots\ldots + x^4\right)^4 \to x^3$

$\left(1 - x^5\right)^4 (1 - x)^{-4} \to x^3$

$\therefore {}^{4+3-1}C_3 = {}^6C_3 = 20$

If sum = 5

$\left(1 - 4x^5\right)(1 - x)^{-4} \to x^5$

$\Rightarrow {}^{4+5-1}C_5 - 4x\ {}^{4.4+0-1}C_0 = {}^8C_5 - 4 = 52$

If sum = 7

$\left(1 - 4x^5\right)(1 - x)^{-4} \to x^7$

$\Rightarrow {}^{4+5-1}C_4 - {}^{4.4+0-1}C_0 = {}^8C_5 - 4 = 52$

If sum = 11

$\left(1 - 4x^5 + 6x^{10}\right)(1 - x)^{-4} \to x^{11}$

$\Rightarrow {}^{4+11-1}C_{11} - 4.\ {}^{4+6-4}C_6 + 6.\ {}^{4+1-1}C_1$

$= {}^{14}C_{11} - 4.{}^{9}C_6 + 6.4 = 364 - 336 + 24 = 52$

$\therefore$ Total matrices = 20 + 52 + 80 + 52 = 204

164. Let $A = \begin{bmatrix} 0 & 1 & 2 \\ a & 0 & 3 \\ 1 & c & 0 \end{bmatrix}$, where $a, c \in \mathbb{R}$. If $A^3 = A$ and the positive value of a belongs to the interval $(n-1, n]$, where $n \in \mathbb{N}$, then n is equal to _____.

माना $A = \begin{bmatrix} 0 & 1 & 2 \\ a & 0 & 3 \\ 1 & c & 0 \end{bmatrix}$, $a, c \in \mathbb{R}$ है। यदि $A^3 = A$ है तथा a का घनात्मक मान अंतराल $(n-1, n]$ में है, जहाँ $n \in \mathbb{N}$ है, तो n बराबर है।

Ans. 2

Sol. $A^2 = \begin{bmatrix} 0 & 1 & 2 \\ a & 0 & 3 \\ 1 & c & 0 \end{bmatrix}\begin{bmatrix} 0 & 1 & 2 \\ a & 0 & 3 \\ 1 & c & 0 \end{bmatrix}$

$= \begin{bmatrix} a+2 & 2c & 3 \\ 3 & a+3c & 2a \\ ca & 1 & 2+3c \end{bmatrix}$

$A^3 = \begin{bmatrix} a+2 & 2c & 3 \\ 3 & a+3c & 2a \\ ca & 1 & 2+3c \end{bmatrix}\begin{bmatrix} 0 & 1 & 2 \\ a & 0 & 3 \\ 1 & c & 0 \end{bmatrix}$

$= \begin{bmatrix} 2ca+3 & a+2+3c & 2a+4+6c \\ a^2+3ca+2a & 3+2ac & 6+3a+9c \\ a+2+3c & ca+2c+3c^2 & 2ca+3 \end{bmatrix}$

$2ca + 3 = 0, a + 2 + 3c = 1$

$a + 1 + 3\left(\frac{-3}{2a}\right) = 1$

$a + 1 - \frac{9}{2a} = 0$

$2a^2 + 2a - 9 = 0$

$a \in (1, 2]$